Executive Summary

Project Title: Agentic Sentinel: Autonomous Trust Layers for Hybrid Quantum-Classical Architectures

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Target Infrastructure: IBM Heron r2 / Enterprise Financial Cores

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As enterprise digital cores modernize, they face two simultaneous, catastrophic vectors of disruption: the acceleration of autonomous Agentic AI workflows, and the cryptographic vulnerabilities exposed by advanced quantum algorithms. Current legacy systems rely on RSA and AES encryption protocols—frameworks that are mathematically guaranteed to fail against the exponential speedup of Shor’s Algorithm and the massive amplitude siphoning of Grover’s Algorithm.

To secure the financial architectures of tomorrow, we must deploy defense mechanisms that operate at the same speed and scale as the threats.

The **Agentic Sentinel** project proposes a high-throughput, Hybrid Quantum-Classical governance model. Designed to run in tandem with processors like the IBM Heron r2, Sentinel acts as the "Classical Optimizer" in a Variational Quantum Eigensolver (VQE) framework. It establishes an automated **Trust Layer** that actively governs high-speed AI agents, ensuring they do not execute vulnerable code or inject cryptographic weaknesses during legacy-to-cloud migrations.

Core Architectural Pillars:

1. **Lattice-Based Post-Quantum Cryptography (PQC):** Transitioning enterprise workflows from standard prime-factorization defenses to multi-dimensional PQC lattices, intentionally injecting "Learning With Errors" noise to force quantum-assisted adversaries into rapid decoherence.
2. **Algorithmic State Governance:** Enforcing strict T_1 and T_2 coherence monitoring on automated AI workflows. Sentinel guarantees that autonomous agents operate within perfectly isolated computational environments, preventing data leaks or external state interference before the final output measurement.
3. **Active Reset & Throughput Optimization:** Utilizing high-speed Active Qubit Reset protocols to clear chaotic data states instantly after calculation, allowing enterprise systems to maintain massive transaction throughput without suffering from quantum crosstalk or thermal degradation.

By fusing the raw computational exploration of quantum hardware with the strict, cold logic of classical AI governance, Agentic Sentinel ensures that modern financial institutions can adopt autonomous AI securely, fully insulated against the quantum decryption landscape of the next decade.